

LOGISTICS & FLEET PERFORMANCE ANALYTICS

End-to-End Power BI Portfolio Case Study

Revenue • Customers • Delivery • Routes • Drivers • Fleet • Fuel • Maintenance • Safety

Project scope: 14 operational datasets | 85,410 loads | approximately \$262.53M revenue

1. Project Overview

This project is an end-to-end Power BI analytics solution designed to evaluate the financial and operational performance of a trucking and logistics company. The report integrates multiple operational datasets into a unified analytical model and presents management-level KPIs with drill-down views across customers, routes, facilities, drivers, trucks, fuel, maintenance and safety.

Business problem

Logistics data is often distributed across dispatch, delivery, driver, vehicle, fuel, maintenance and safety systems. Analyzing these sources separately makes it difficult to see how commercial and operational performance connect. This project consolidates those sources into one interactive Power BI report.

Primary objectives

Measure revenue and load volume; identify high-value customers; evaluate on-time delivery and detention; compare route and facility performance; analyze driver productivity; monitor fleet utilization and fuel efficiency; quantify maintenance and downtime; and assess safety frequency and financial exposure.

Dataset

The model uses 14 CSV datasets covering customers, delivery events, driver monthly metrics, drivers, facilities, fuel purchases, loads, maintenance records, routes, safety incidents, trailers, trips, truck utilization metrics and trucks.

Loads	85,410	Trips	85,410
Delivery events	170,820	Fuel purchases	196,442
Customers	200	Drivers	150
Trucks	120	Routes	58
Facilities	50	Safety incidents	170

Data preparation & modeling

Power Query was used to standardize IDs, dates, timestamps and numeric fields; trim and clean text; profile column quality; and correct date fields containing hidden time components. The model follows a dimensional approach with dimensions generally filtering fact tables through one-to-many, single-direction relationships. A dedicated Calendar table supports consistent time analysis.

Delivery-event reporting uses the actual event date through an inactive Calendar relationship activated in DAX with USERELATIONSHIP(), avoiding an ambiguous active filter path.

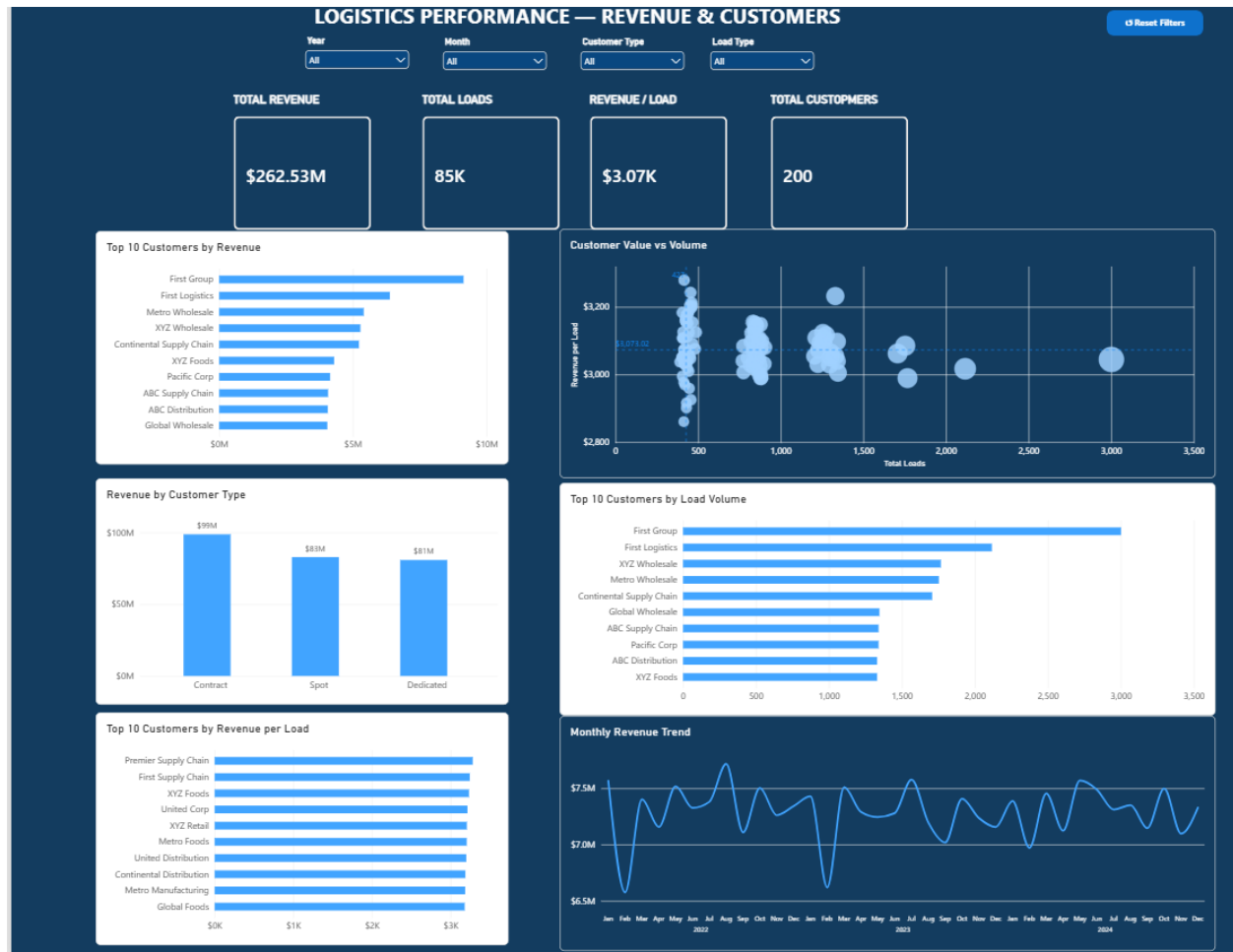
2. Executive Overview



\$262.53M revenue from 85,410 loads, with approximately \$3.07K revenue per load.

- Fleet utilization averages approximately 83.0% and fleet fuel efficiency is approximately 6.45 MPG.
- On-time delivery is 44.6%, making delivery reliability the clearest operational area requiring deeper investigation.
- The page combines commercial, delivery and fleet KPIs with route, customer and monthly trend views.

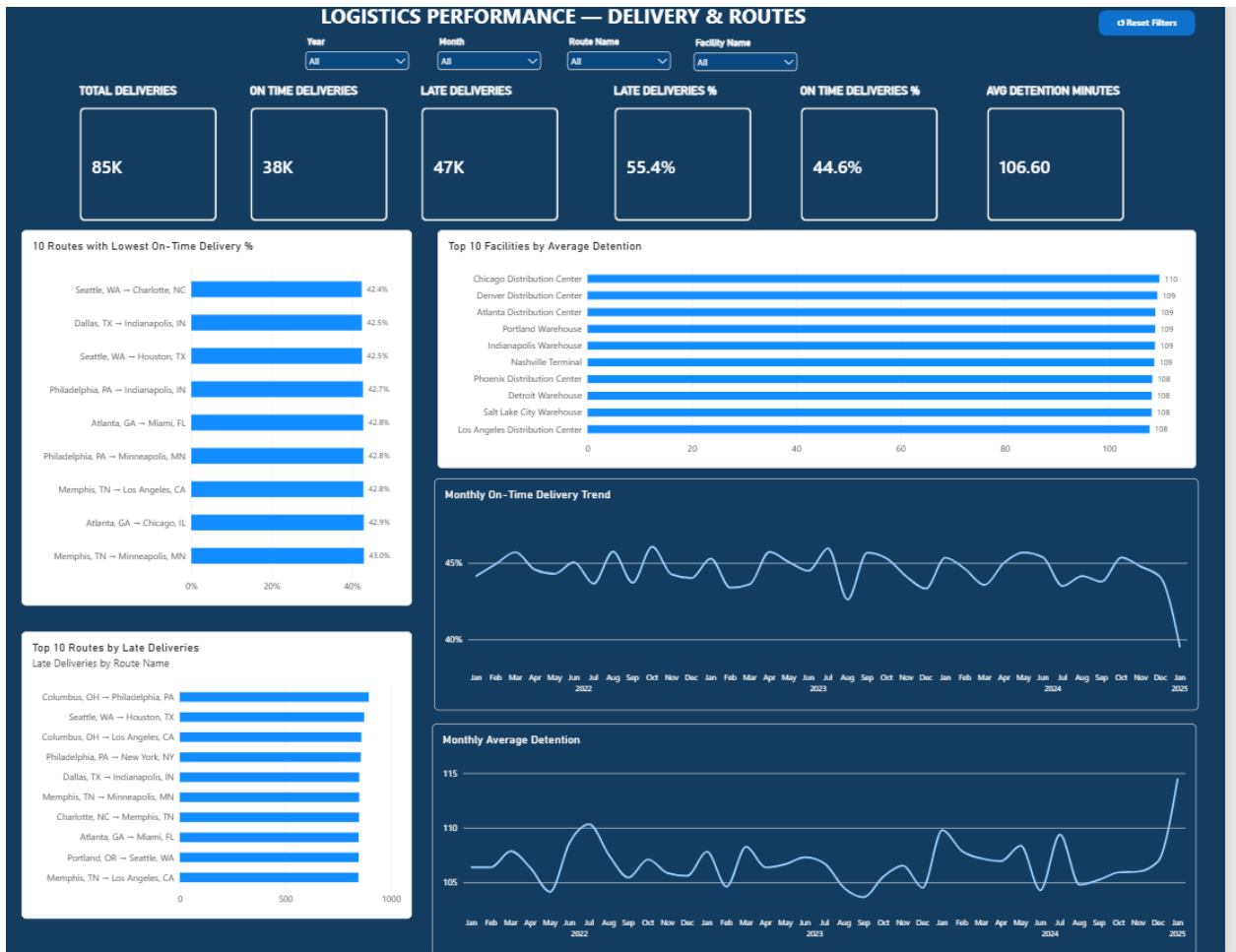
3. Revenue & Customers



Customer performance should be evaluated using revenue, load volume and revenue per load together.

- Contract customers represent the largest customer-type revenue segment at approximately \$99M; Spot and Dedicated contribute approximately \$83M and \$81M respectively.
- The value-versus-volume view separates large-volume accounts from accounts producing stronger revenue per load.
- This supports account prioritization without assuming that the highest-volume customer is automatically the most attractive on a per-load basis.

4. Delivery & Routes



Only 44.6% of deliveries were on time; approximately 55.4% were late.

- The dataset contains 38,102 on-time deliveries and 47,308 late deliveries.
- Average delivery detention is approximately 106.6 minutes, or about 1 hour 47 minutes.
- Route and facility visuals isolate low on-time routes, routes with many late deliveries and facilities with high detention.
- The dashboard identifies patterns for investigation; it does not establish that detention itself caused late delivery.

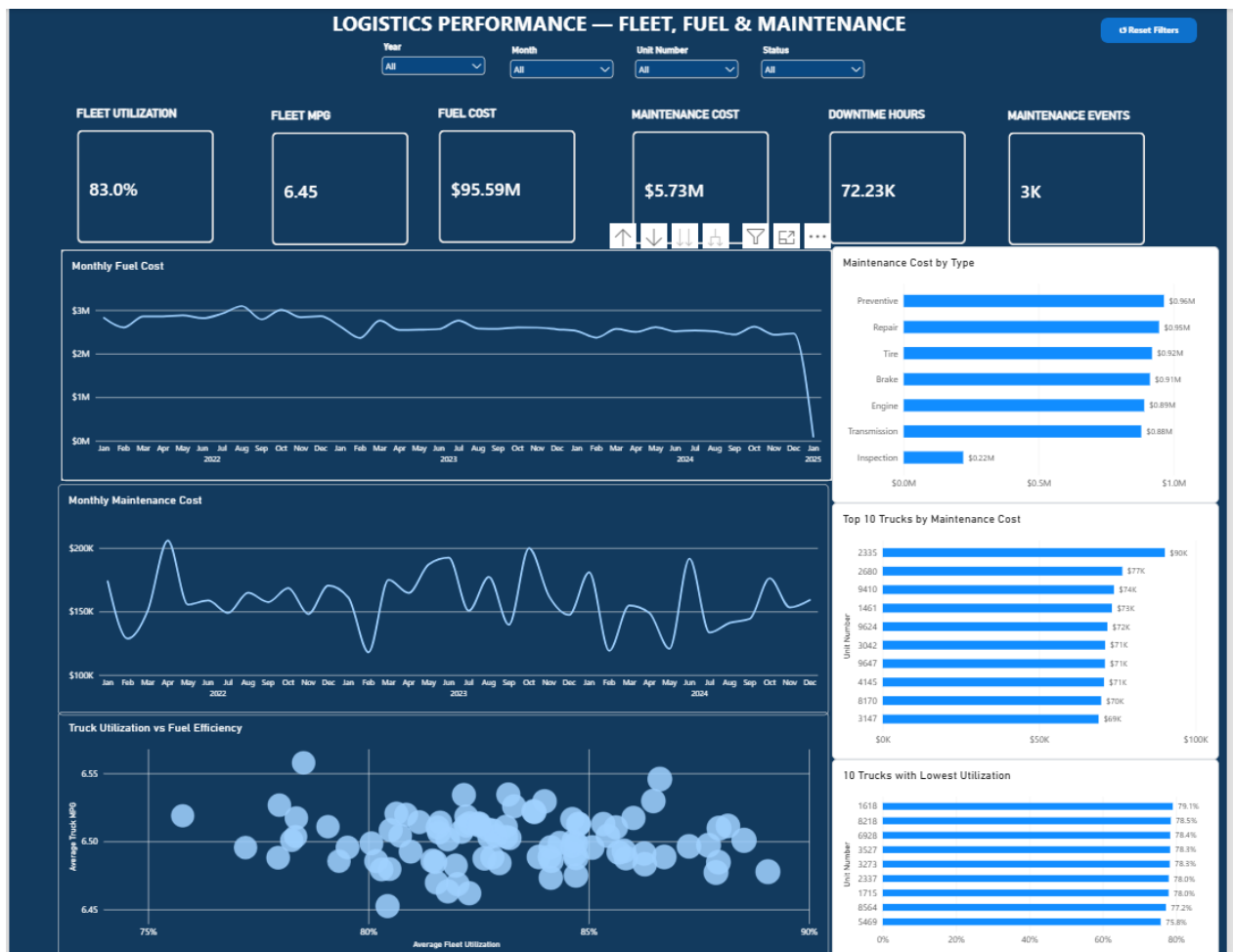
5. Driver Performance



124 drivers recorded activity, producing approximately \$257.26M in driver-attributed revenue.

- Driver metrics include approximately 120M miles, 84K trips, \$2.15 revenue per mile and about 1.4K miles per trip.
- Monthly driver MPG remains relatively stable around the mid-6.4 range.
- The productivity scatterplot shows meaningful differences in total mileage while revenue per mile is comparatively concentrated.
- Mileage differences should be investigated alongside route assignment, working period, availability and load mix.

6. Fleet, Fuel & Maintenance



Fuel expenditure is approximately \$95.59M; maintenance expenditure is approximately \$5.73M.

- Fleet utilization averages approximately 83.0% and average truck MPG is approximately 6.45.
- Recorded maintenance downtime totals approximately 72.23K hours across roughly 3K maintenance events.
- Truck-level visuals identify vehicles with relatively high maintenance costs and low utilization for further asset review.
- The small 2025 fuel point represents partial/spillover data and should not be treated as a complete-year comparison.

7. Safety & Risk



170 safety incidents generated approximately \$2.65M in recorded claims.

- 64 incidents were marked preventable (37.6%), 54 were at-fault and 33 involved injuries.
- DOT Violation was the most frequent recorded incident type (39), while Equipment Damage showed the largest claims amount at approximately \$0.74M.
- Frequency and financial severity should be reviewed together rather than using incident count alone.
- The dashboard identifies drivers and trucks with the most recorded incidents for further investigation.

8. Key Findings & Business Recommendations

Delivery reliability

Finding: 44.6% on-time delivery and 55.4% late delivery, alongside average detention of 106.6 minutes.

Recommendation: Prioritize high-volume routes with weak on-time performance, then examine associated facilities, scheduling patterns and detention.

Customer economics

Finding: Customer value varies by both volume and revenue per load.

Recommendation: Use revenue, load volume and revenue per load together when evaluating account value and commercial priorities.

Driver productivity

Finding: Driver mileage varies substantially while revenue per mile is relatively concentrated.

Recommendation: Investigate route assignment, working period, availability and load mix before interpreting mileage differences as individual performance.

Fleet economics

Finding: Fuel is a major recorded operating-cost category, while truck-level maintenance cost and utilization vary.

Recommendation: Review trucks that combine low utilization, high maintenance expense or excessive downtime as candidates for deeper repair-versus-replacement analysis.

Safety exposure

Finding: 37.6% of incidents are marked preventable and incident frequency does not perfectly align with claims severity.

Recommendation: Prioritize safety interventions using frequency, preventability, fault, injury occurrence and claims exposure together.

Management takeaway

The strongest cross-functional issue in the report is delivery reliability. The dashboard provides route- and facility-level views needed to move from the company-wide KPI toward targeted operational investigation while preserving commercial, fleet and safety context.

9. Data Quality, Limitations & Skills Demonstrated

Data-quality findings

Fuel attribution: 3,880 fuel-purchase records have no truck identifier. They remain included in fleet-wide fuel totals but cannot be attributed reliably to an individual truck.

Safety attribution: 1 of 170 safety incidents has no driver identifier. It remains included in company-level safety metrics but cannot be attributed to an individual driver.

Partial 2025 period: Delivery-event data extends into 2025, but only 86 delivery events occur in that year. It is treated as a partial/spillover period rather than a full-year comparison.

Analytical limitations

The dashboard identifies associations and operational patterns; it does not by itself establish causal relationships. For example, facilities with high detention may coincide with delivery performance issues, but additional analysis would be required to establish whether detention caused specific late deliveries.

Skills demonstrated

Power BI Desktop	Power Query	DAX
Data Cleaning	Dimensional Modeling	Star-Schema Concepts
Time Intelligence	KPI Development	Data Visualization
Data Quality Assessment	Business Analysis	Dashboard Design

Portfolio summary

Built an end-to-end Power BI solution integrating 14 logistics datasets covering 85K+ loads and \$262M+ in revenue. Developed Power Query transformations, a dimensional model and DAX KPIs across revenue, delivery, driver, fleet, fuel, maintenance and safety performance, while documenting material data-quality limitations instead of hiding them.